endurance, multi-role UAVs in both military and non-military domains. These systems demonstrated how advanced autonomy, endurance, and precision strike capability could redefine reconnaissance, surveillance, and warfare strategies across the globe. The following case studies highlight the operational significance, strategic outcomes, and technological impact of key third-generation UAVs in real-world missions.

### 4.7.1 Case study 1: RQ-4 Global Hawk – Strategic intelligence in global surveillance (1998–Present)

The RQ-4 Global Hawk, developed by Northrop Grumman (United States), became a cornerstone of global intelligence operations for the US Air Force and allied forces. Its ability to fly at altitudes above 60,000 feet for over 36 hours allowed persistent surveillance over vast regions, including conflict zones and maritime areas. During operations in Afghanistan, Iraq, and Libya, the Global Hawk provided real-time strategic imagery, signals intelligence (SIGINT), and battlefield mapping to support mission planning and target assessment. Beyond military applications, the RQ-4 was instrumental in humanitarian and disaster-response operations, such as post-tsunami mapping in Japan (2011) and wildfire surveillance in California. The drone’s autonomous navigation and multi-sensor integration demonstrated the feasibility of continuous, high-altitude, global ISR missions. Its success established the foundation for future HALE UAV programs focused on global reconnaissance and autonomous data fusion.

### 4.7.2 Case study 2: MQ-9 Reaper – Precision strike and counterterrorism operations (2007–Present)

The MQ-9 Reaper, developed by General Atomics Aeronautical Systems (United States), revolutionized unmanned combat with its ability to perform simultaneous reconnaissance and precision strike missions. It became a